Report Exploration Bertoldi

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Cohort description:

A study is currently being conducted involving patients affected by multiple sclerosis, recruited from different clinical centers. Specifically, patients are affiliated with the Clerico and CReSM centers at San Luigi Gonzaga Hospital in Turin, the Molinette Hospital in Turin, as well as centers in Brescia and Naples.

```
table(merged$CENTRO)
```

```
##
##   BRESCIA   CLERICO     CRESM MOLINETTE    NAPOLI
##         2        43         6         3         8
```

The overall cohort includes 61 patients, of whom 31 are treated with dimethyl fumarate (Tecfidera) and 30 with cladribine (Mavenclad).

```
table(merged$therapy)
```

```
##
##        cladribine dimethyl_fumarate
##                30                32
```

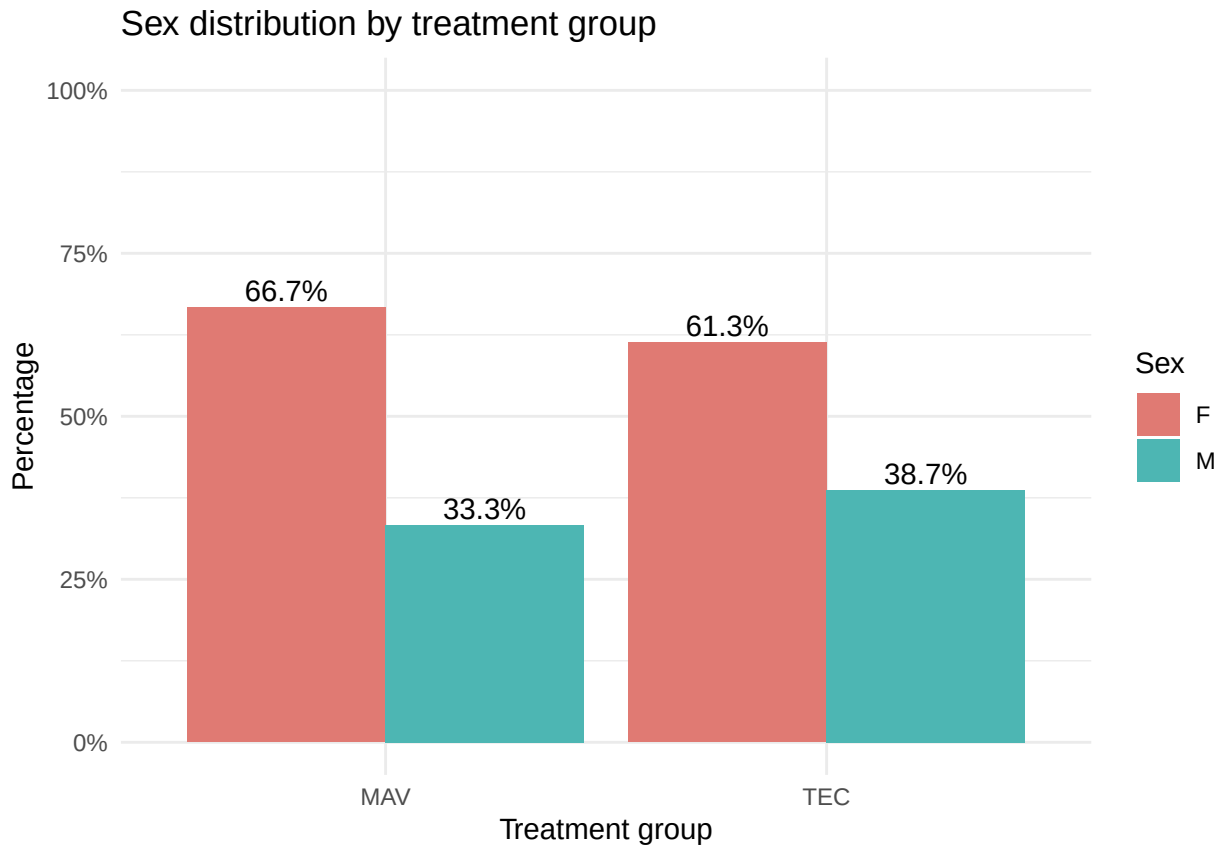
Regarding sex distribution, 21 patients are male and 39 are female, showing a predominance of females within the cohort.

```
table(merged$sex, useNA = "ifany")
```

```
##
##    F    M <NA>
##   39   22    1
```

```
#% of sex distribution
ggplot(sex_prop, aes(x = GROUP, y = percent, fill = sex)) +
geom_col(position = "dodge") +
  geom_text(
    aes(label = scales::percent(percent)),
    position = position_dodge(width = 0.9),
    vjust = -0.3,
    size = 4
  ) +
```

```
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  scale_fill_manual(values = c(
    "F" = "#E07A73",
    "M" = "#4DB6B3"
  )) +
  labs(
    title = "Sex distribution by treatment group",
    x = "Treatment group",
    y = "Percentage",
    fill = "Sex"
  ) +
  theme_minimal()
```



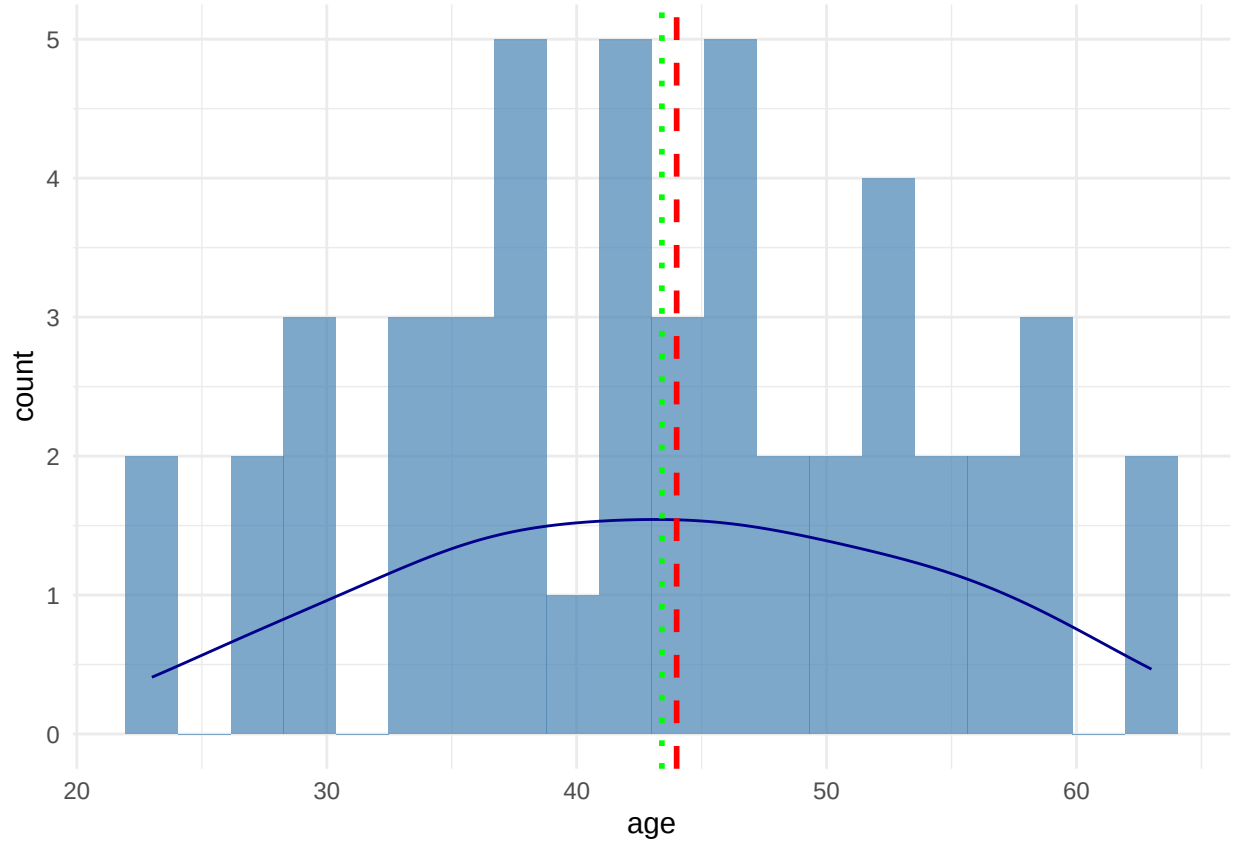
Age distribution in the study cohort The age distribution at treatment initiation is approximately symmetrical, with most patients between 35 and 55 years. Although Tecfidera patients are slightly older on average, the difference is not statistically significant ($p = 0.19$), and both groups show comparable variability and age range, reflecting a representative adult multiple sclerosis population.

```
#age distribution
ggplot(merged, aes(x = age)) +
  geom_histogram(bins = 20, fill = "steelblue", alpha = 0.7) +
  geom_density(aes(y = after_stat(count)), color = "darkblue") +
  geom_vline(aes(xintercept = median(age, na.rm = TRUE)),
             color = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(aes(xintercept = mean(age, na.rm = TRUE)),
             color = "green", linetype = "dotted", linewidth = 1) +
```

```
  theme_minimal()
```

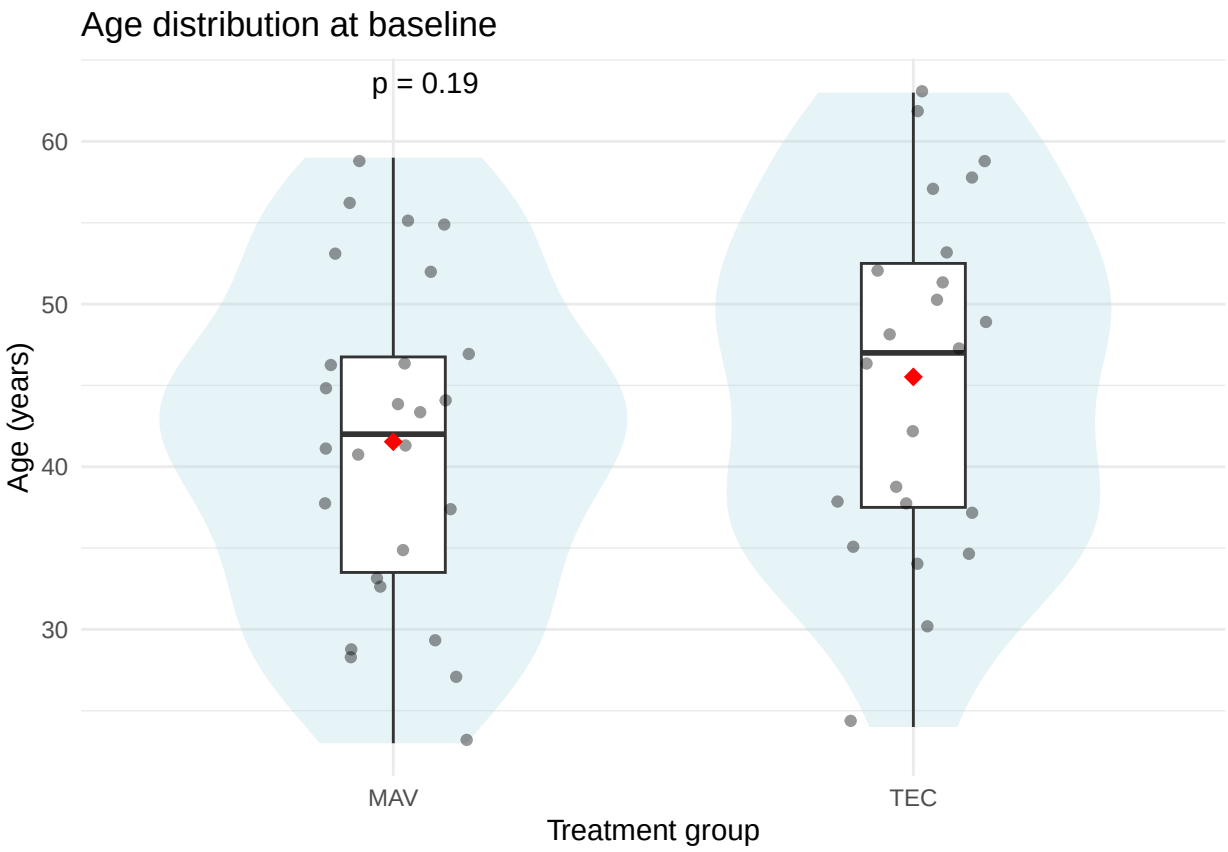
```
## Warning: Removed 13 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

```
## Warning: Removed 13 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```
#age distribution between TEC and MAV
ggplot(merged %>% filter(!is.na(age)), aes(x = GROUP, y = age)) +
  geom_violin(fill = "lightblue", alpha = 0.3, color = NA) +
  geom_boxplot(width = 0.2, outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.4) +
  stat_summary(
    fun = mean,
    geom = "point",
    shape = 18,
    size = 3,
    color = "red"
  ) +
  stat_compare_means(
    method = "wilcox.test",
    label = "p.format"
  ) +
```

```
  theme_minimal() +
  labs(
    title = "Age distribution at baseline",
    x = "Treatment group",
    y = "Age (years)"
  )
```



Age at diagnosis in the study cohort

Age at diagnosis was analyzed by treatment group (MAV vs TEC). Both groups show wide and overlapping distributions, with a slight trend toward higher values in the TEC group. Mean age at diagnosis was 38.3 years (median 37, SD = 12.0) in the TEC group and 34.4 years (median 32, SD = 10.3) in the MAV group. Variability was similar across groups, though slightly higher in TEC.

```
#TEC at diagnosis
merged %>%
  filter(str_starts(ID, "TEC")) %>%
  summarise(
    n = n(),
    eta_media = mean(age_at_diagnosis, na.rm = TRUE),
    eta_mediana = median(age_at_diagnosis, na.rm = TRUE),
    sd = sd(age_at_diagnosis, na.rm = TRUE)
  )
```

```
##    n eta_media eta_mediana       sd
```

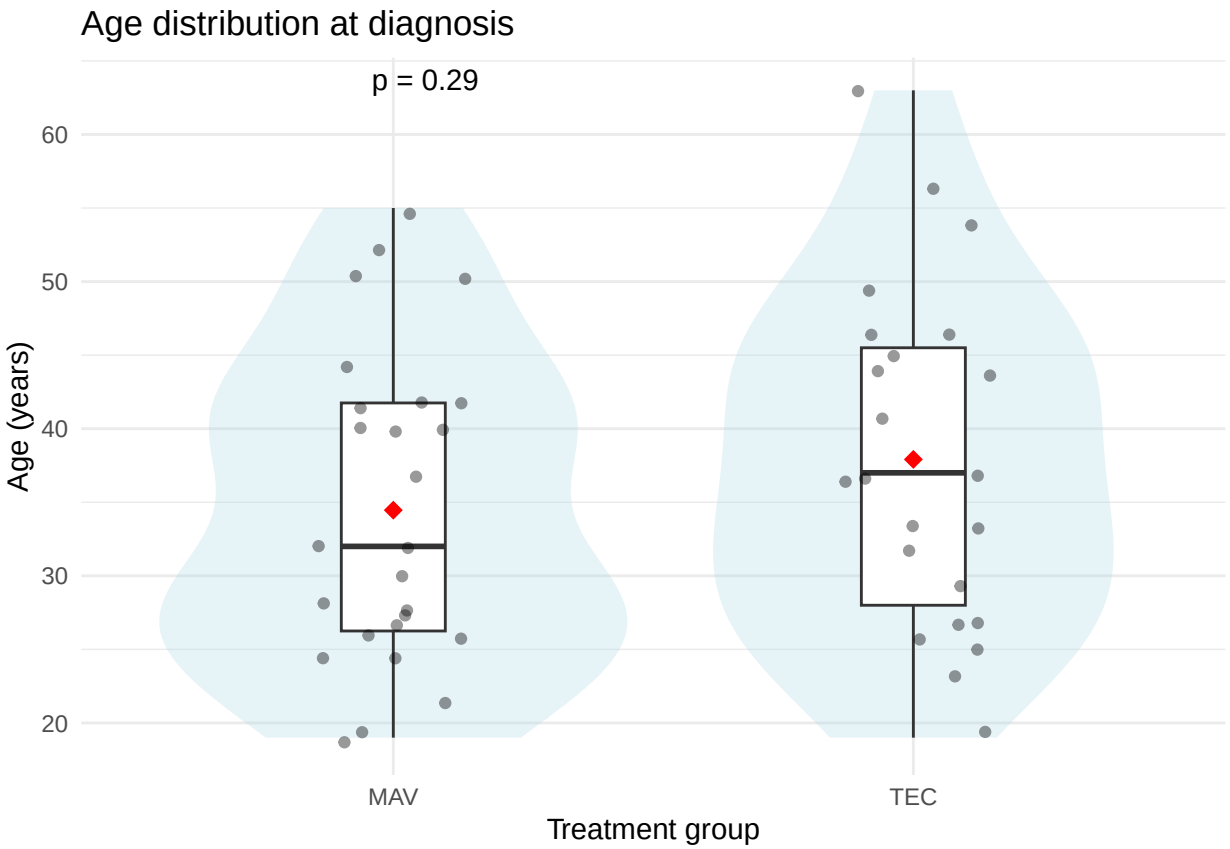
```
## 1 32  37.91304          37 11.47691
```

```r
#MAV at diagnosis
merged %>%
  filter(str_starts(ID, "MAV")) %>%
  summarise(
    n = n(),
    eta_media = mean(age_at_diagnosis, na.rm = TRUE),
    eta_mediana = median(age_at_diagnosis, na.rm = TRUE),
    sd = sd(age_at_diagnosis, na.rm = TRUE)
  )
```

```
##    n eta_media eta_mediana       sd
## 1 30  34.46154          32 10.59521
```

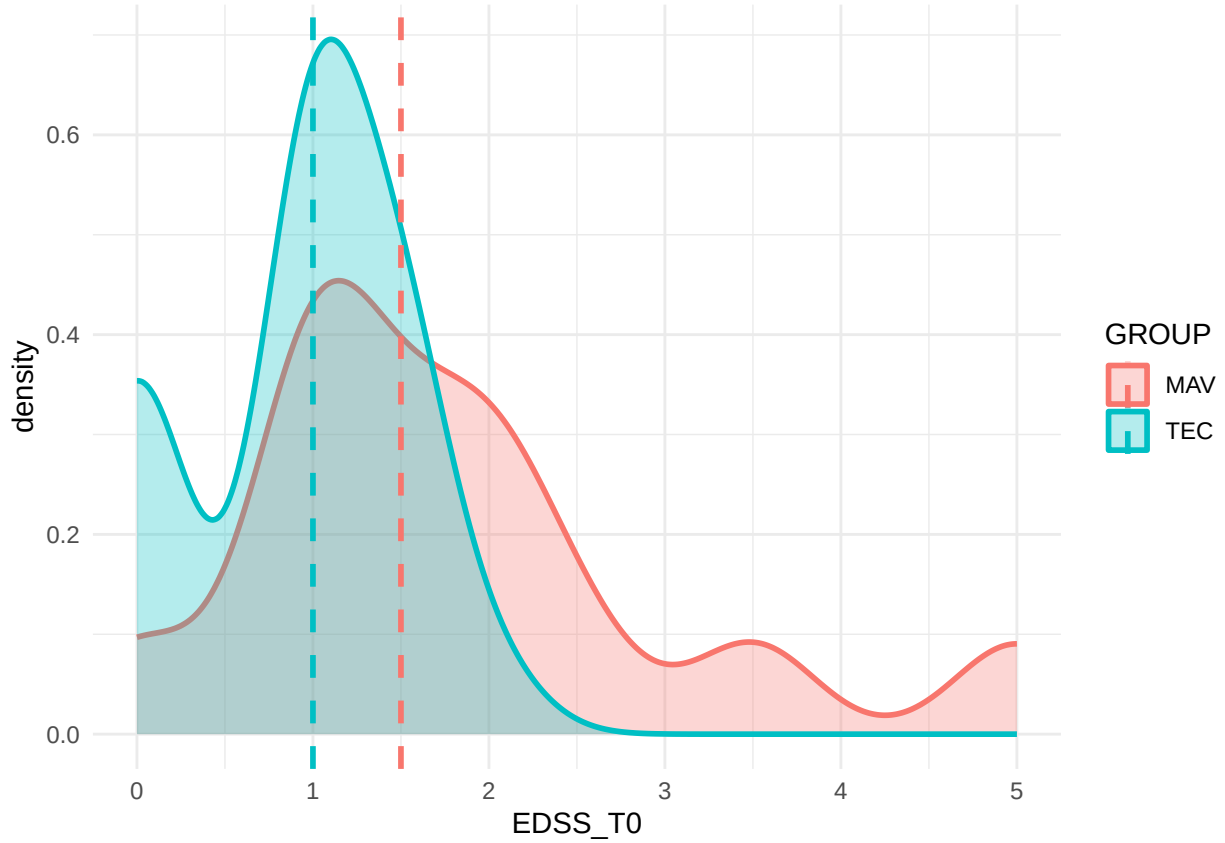
No statistically significant difference was observed between the two groups ($p = 0.2$), indicating comparable age at diagnosis.

```r
#Age at diagnosis by treatment group
ggplot(merged %>% filter(!is.na(age_at_diagnosis)), aes(x = GROUP, y = age_at_diagnosis)) +
  geom_violin(fill = "lightblue", alpha = 0.3, color = NA) +
  geom_boxplot(width = 0.2, outlier.shape = NA) +
  geom_jitter(width = 0.15, alpha = 0.4) +
  stat_summary(
    fun = mean,
    geom = "point",
    shape = 18,
    size = 3,
    color = "red"
  ) +
  stat_compare_means(
    method = "wilcox.test",
    label = "p.format"
  ) +
  theme_minimal() +
  labs(
    title = "Age distribution at diagnosis",
    x = "Treatment group",
    y = "Age (years)"
  )
```

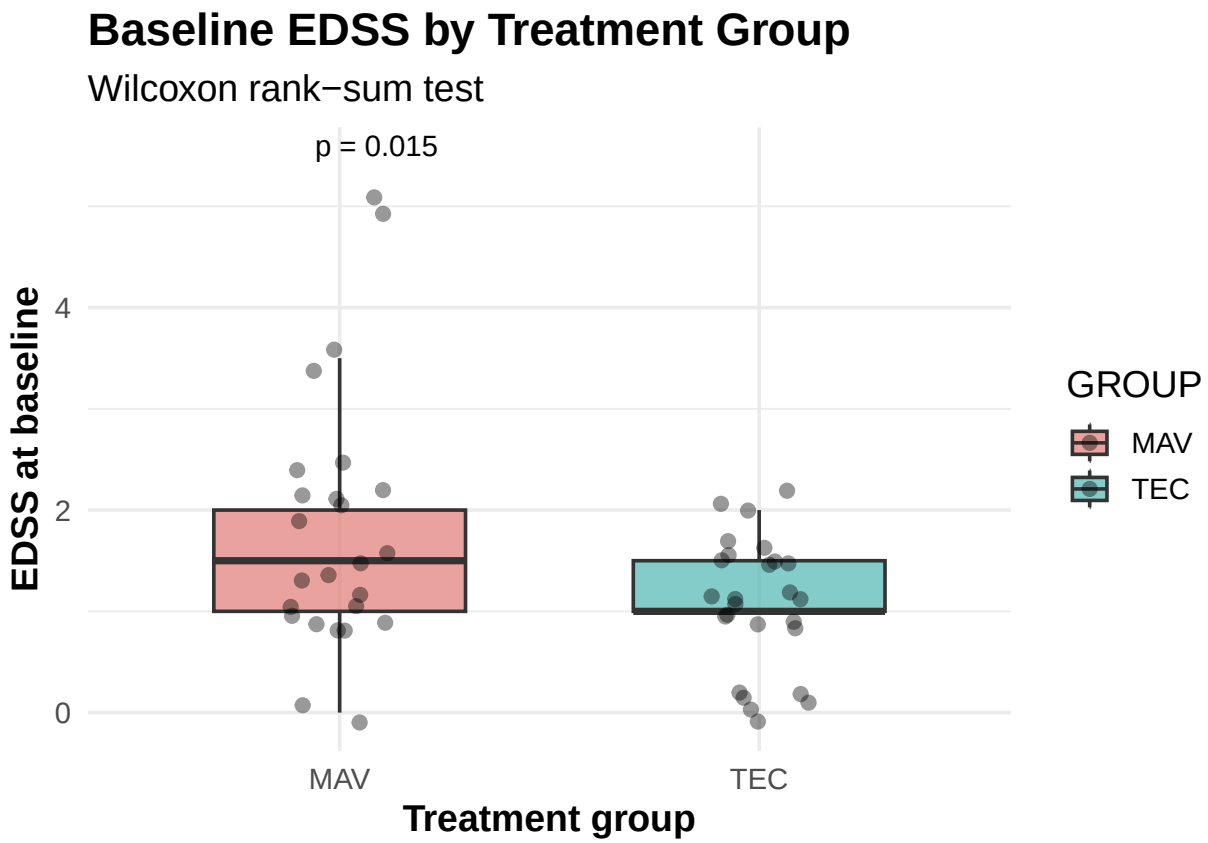


##Baseline EDSS focus At baseline, EDSS values were predominantly within the mild–moderate disability range. The distribution differed between treatment groups, with the TEC group showing lower and more concentrated values, while the MAV group appeared more heterogeneous and slightly shifted toward higher EDSS scores. These differences were confirmed by boxplot analysis, which showed a significantly higher baseline EDSS in the MAV group compared to the TEC group (p = 0.006).

```
#density plot
ggplot(
merged %>%
    filter(!is.na(EDSS_T0), !is.na(age_at_diagnosis))
, aes(x = EDSS_T0, color = GROUP, fill = GROUP)) +
  geom_density(alpha = 0.3, linewidth = 1) +
   geom_vline(
    data = stats_EDSS_baseline,
    aes(xintercept = mediana, color = GROUP),
    linetype = "dashed",
    linewidth = 1
  )+
  theme_minimal()
```




```r
#boxplot+jitter plot
ggplot(merged %>% filter(!is.na(EDSS_T0)), aes(x = GROUP, y = EDSS_T0, fill = GROUP)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, width = 0.6) +
  geom_jitter(width = 0.12, alpha = 0.4, size = 2, color = "black") +
  stat_compare_means(
    method = "wilcox.test",
    label = "p.format",
    label.y = max(merged$EDSS_T0, na.rm = TRUE) + 0.5,
    size = 4
  ) +
  scale_fill_manual(values = c("MAV" = "#E07A73", "TEC" = "#4DB6B3")) +
  labs(
    title = "Baseline EDSS by Treatment Group",
    subtitle = "Wilcoxon rank-sum test",
    x = "Treatment group",
    y = "EDSS at baseline"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold"),
    axis.title = element_text(face = "bold")
  )
```



##Longitudinal EDSS trajectories EDSS over time shows moderate interindividual variability, with heterogeneous trajectories generally within a mild–moderate range. Mean EDSS remains largely stable, with a slight increase up to around 18–20 months followed by stabilization, suggesting no clear overall progression.

```
ggplot(edss_long, aes(x = mesi, y = EDSS, group = ID)) +
  geom_line(alpha = 0.25, color = "grey50", linewidth = 0.6) +
  geom_point(alpha = 0.4, size = 1, color = "grey40") +
  stat_summary(
    aes(group = 1),
    fun.data = mean_se,
    geom = "ribbon",
    fill = "red",
    alpha = 0.2,
    color = NA
  ) +
  stat_summary(
    aes(group = 1),
    fun = mean,
    geom = "line",

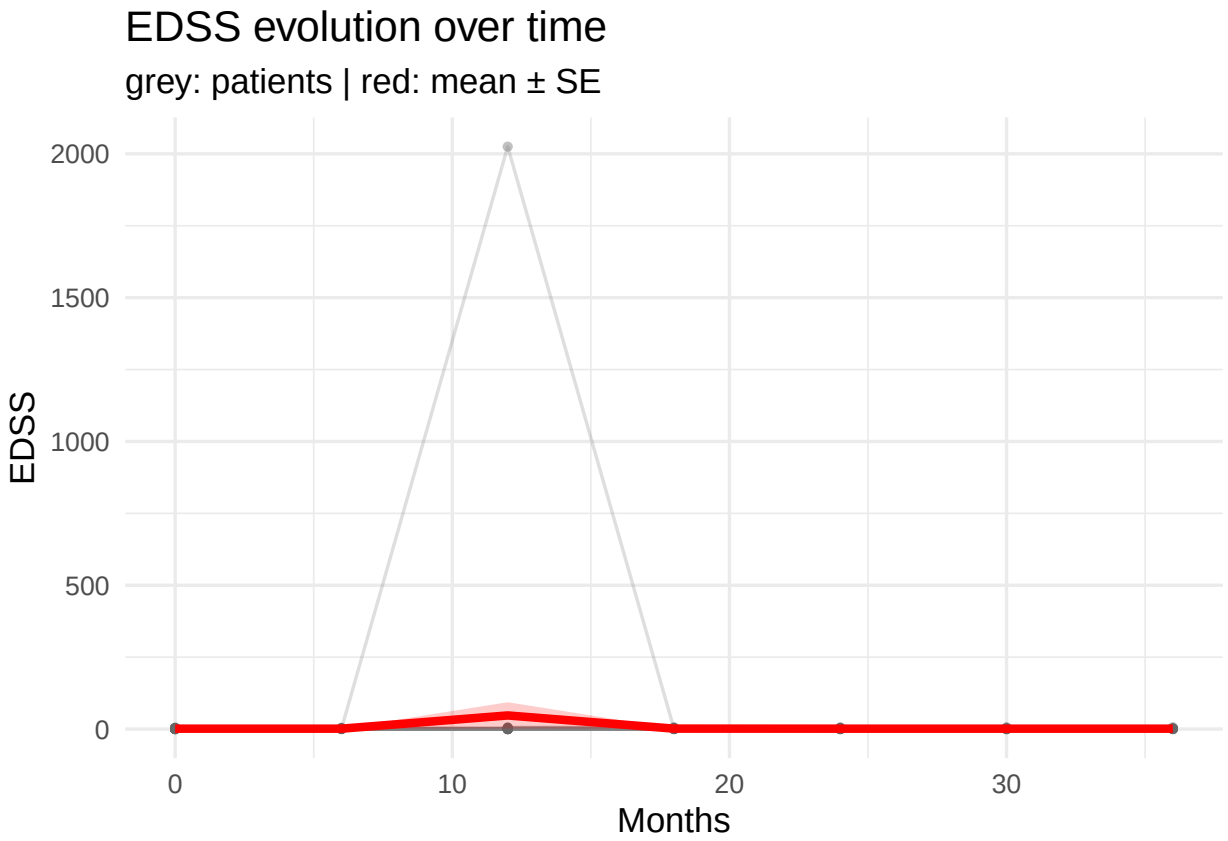
    color = "red",
    linewidth = 1.5
  ) +
  labs(
    title = "EDSS evolution over time",
    subtitle = "grey: patients | red: mean ± SE",
```

```
    x = "Months",
    y = "EDSS"
  ) +
  theme_minimal(base_size = 13)
```

```
## Warning: Removed 227 rows containing non-finite outside the scale range
## (`stat_summary()`).
## Removed 227 rows containing non-finite outside the scale range
## (`stat_summary()`).
```

```
## Warning: Removed 174 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 227 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

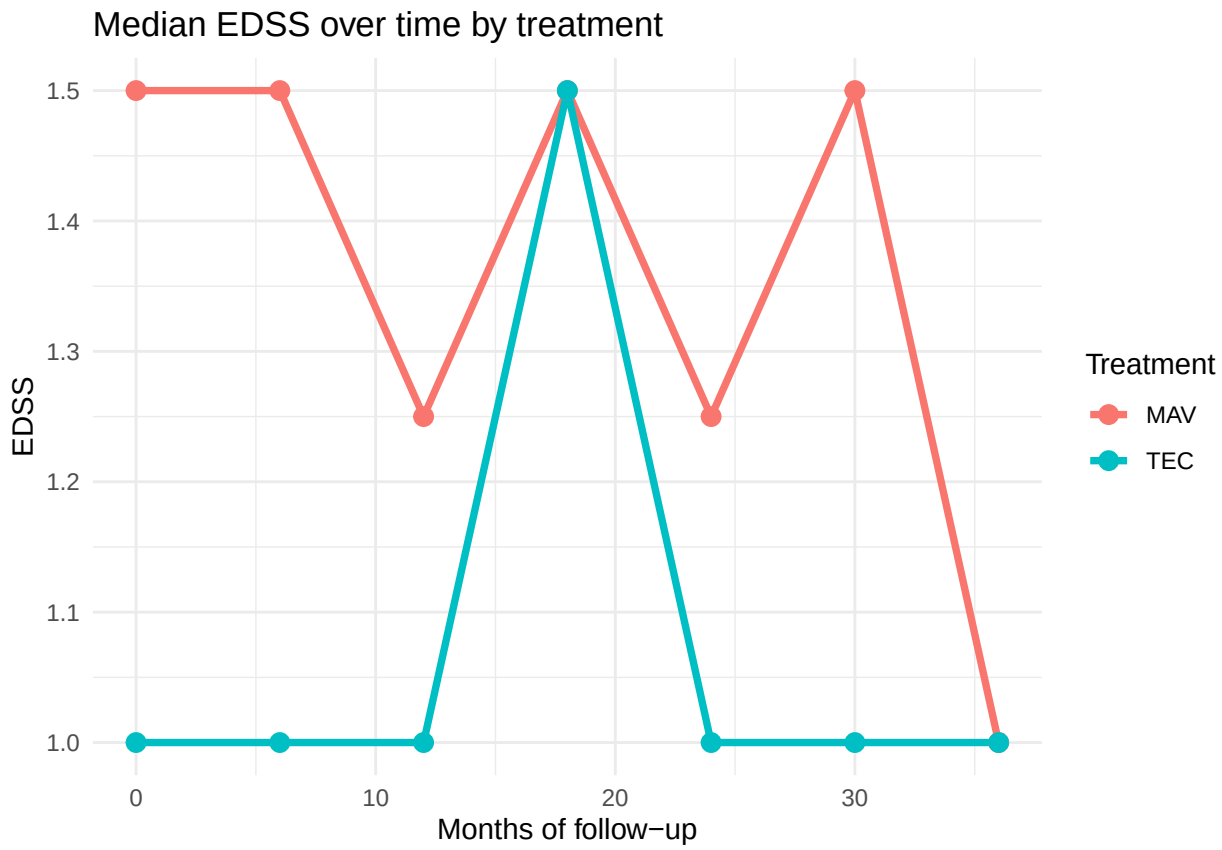


##EDSS trends by treatment group Median EDSS differed between treatment groups over time, with higher baseline values in the MAV group and relatively stable trajectories in both groups, followed by a decrease at later timepoints. Interpretation of late timepoints should be cautious due to reduced sample size.

```
ggplot(edss_long, aes(x = mesi, y = EDSS, color = farmaco)) +
  stat_summary(
    fun = median,
    geom = "line",
```

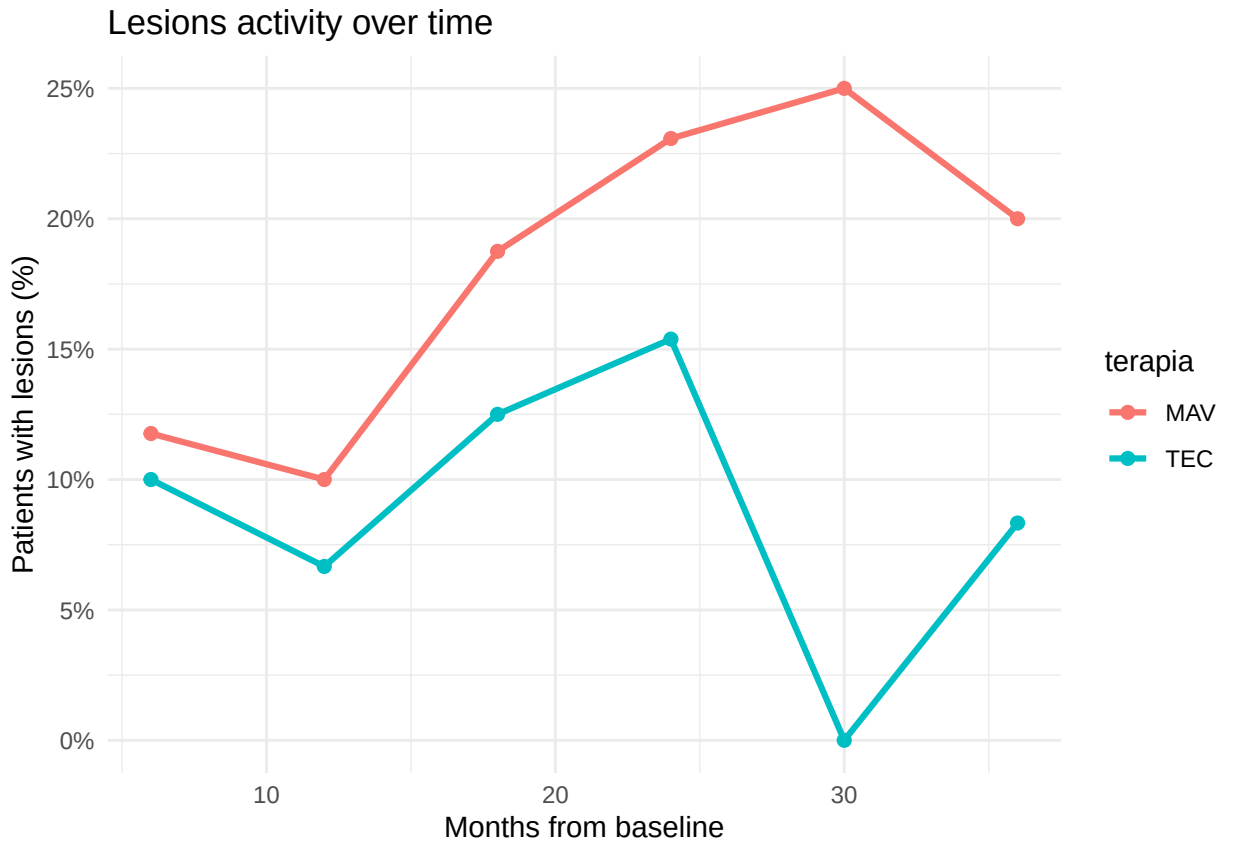
```
    linewidth = 1.3
  ) +
  stat_summary(
    fun = median,
    geom = "point",
    size = 3
  ) +
  labs(
    title = "Median EDSS over time by treatment",
    x = "Months of follow-up",
    y = "EDSS",
    color = "Treatment"
  ) +
  theme_minimal()
```

```
## Warning: Removed 227 rows containing non-finite outside the scale range
## (`stat_summary()`).
## Removed 227 rows containing non-finite outside the scale range
## (`stat_summary()`).
```



##Disease evolution: lesion activity over time by treatment The proportion of patients with new lesions was consistently higher in the MAV group compared to TEC, with a peak at 30 months. Lower activity was observed in the TEC group, although late timepoints should be interpreted with caution due to reduced sample size.

```r
ggplot(lesioni_prop, aes(x = mesi, y = prop_lesioni, color = terapia)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2) +
  scale_y_continuous(labels = scales::percent) +
  labs(
    x = "Months from baseline",
    y = "Patients with lesions (%)",
    title = "Lesions activity over time"
  ) +
  theme_minimal()
```



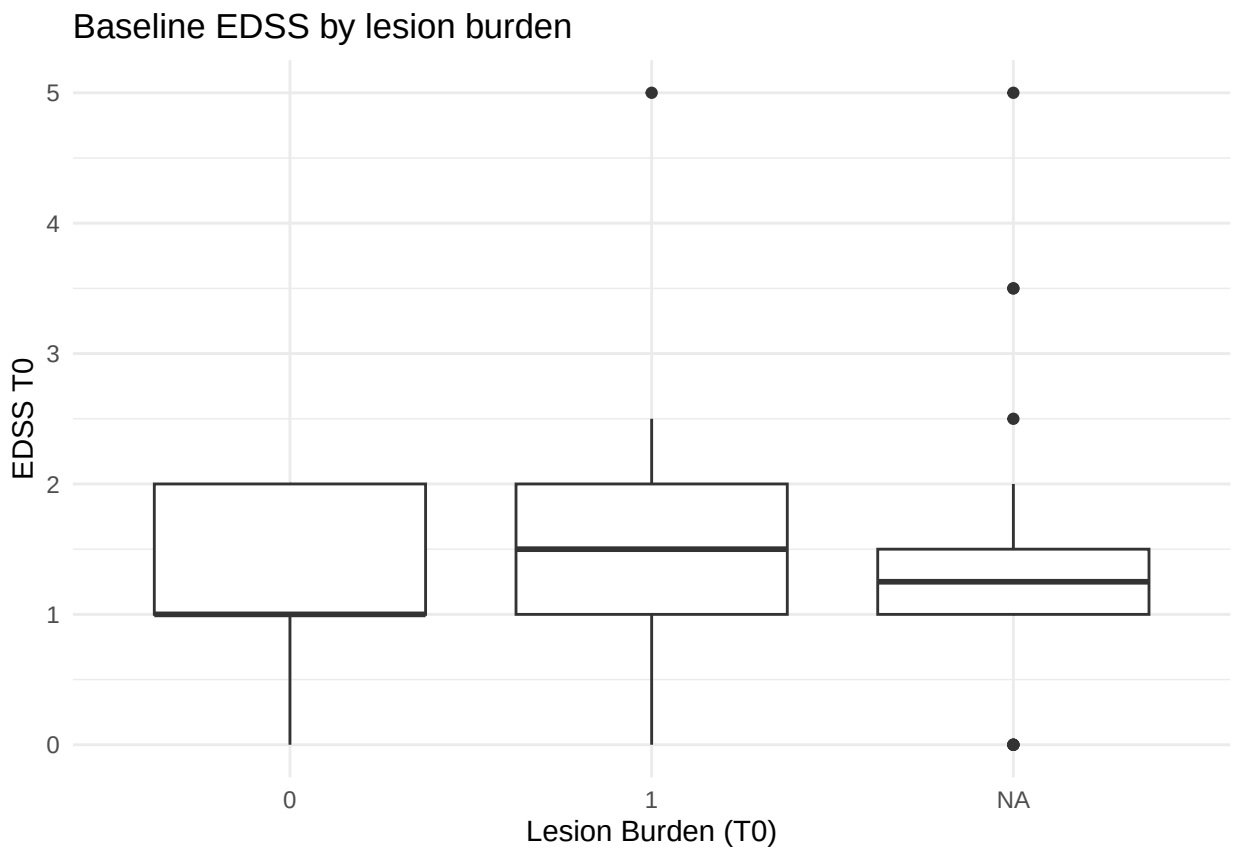
Exploratory analysis: factors associated with baseline EDSS

An exploratory analysis was conducted to assess factors associated with baseline EDSS. A significant association was observed between lesion burden and EDSS, with higher lesion load corresponding to increased disability ($p = 0.015$).

```r
ggplot(merged, aes(x = factor(lesion_burden), y = EDSS_T0)) +
  geom_boxplot() +
  labs(
    x = "Lesion Burden (T0)",
    y = "EDSS T0",
    title = "Baseline EDSS by lesion burden"
```

```
  ) +
  theme_minimal()
```

```
## Warning: Removed 11 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```




```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): non è
## possibile calcolare p-value esatto in presenza di ties
```

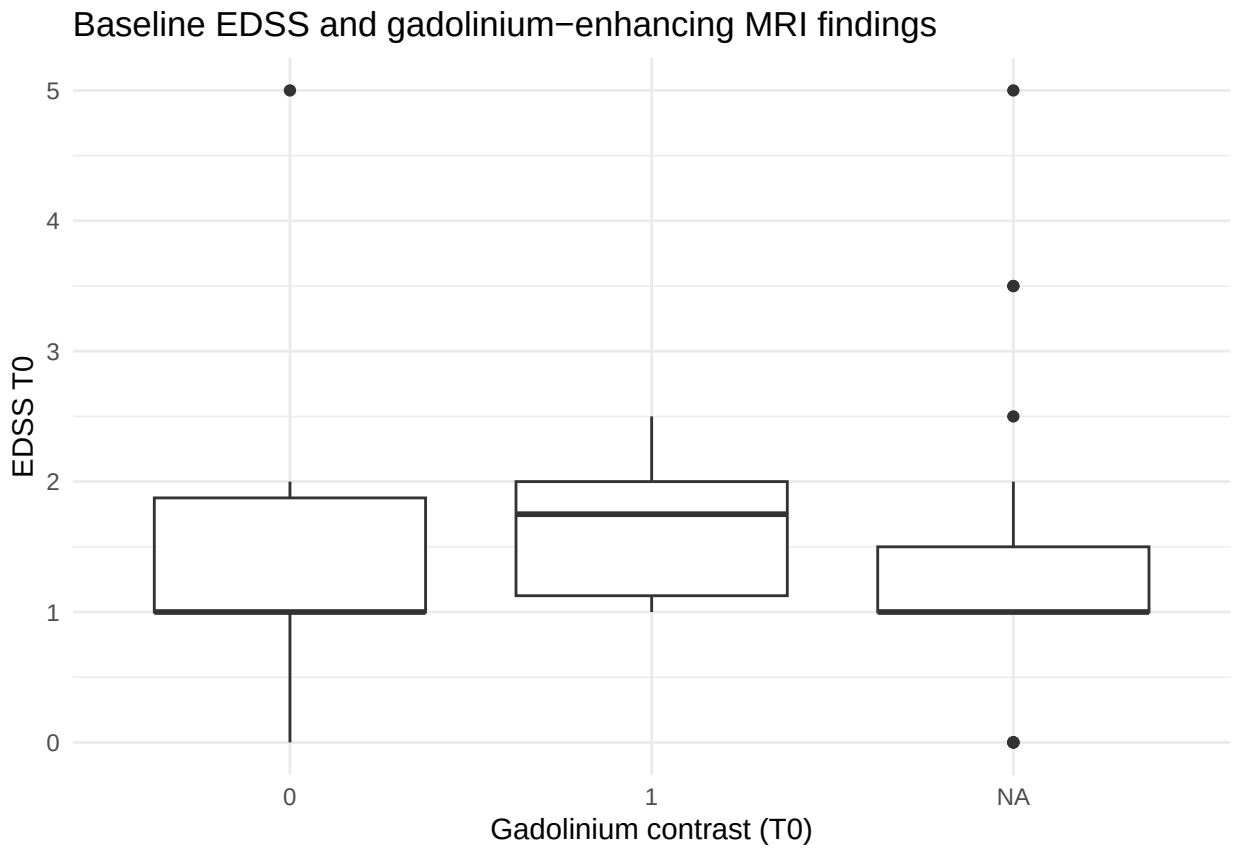
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  EDSS_T0 by lesion_burden
## W = 72.5, p-value = 0.1574
## alternative hypothesis: true location shift is not equal to 0
```

Patients with gadolinium-enhancing lesions showed higher EDSS values, although this difference did not reach statistical significance ($p = 0.165$). No significant correlations were found between EDSS and age at treatment initiation or age at diagnosis.

```
ggplot(merged, aes(x = factor(gadolinium_contrast), y = EDSS_T0)) +
  geom_boxplot() +
  labs(
```

```
    x = "Gadolinium contrast (T0)",
    y = "EDSS T0",
    title = "Baseline EDSS and gadolinium-enhancing MRI findings"
  ) +
  theme_minimal()
```

```
## Warning: Removed 11 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```




```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): non è
## possibile calcolare p-value esatto in presenza di ties
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  EDSS_T0 by gadolinium_contrast
## W = 44, p-value = 0.2123
## alternative hypothesis: true location shift is not equal to 0
```

Smoking status and baseline EDSS

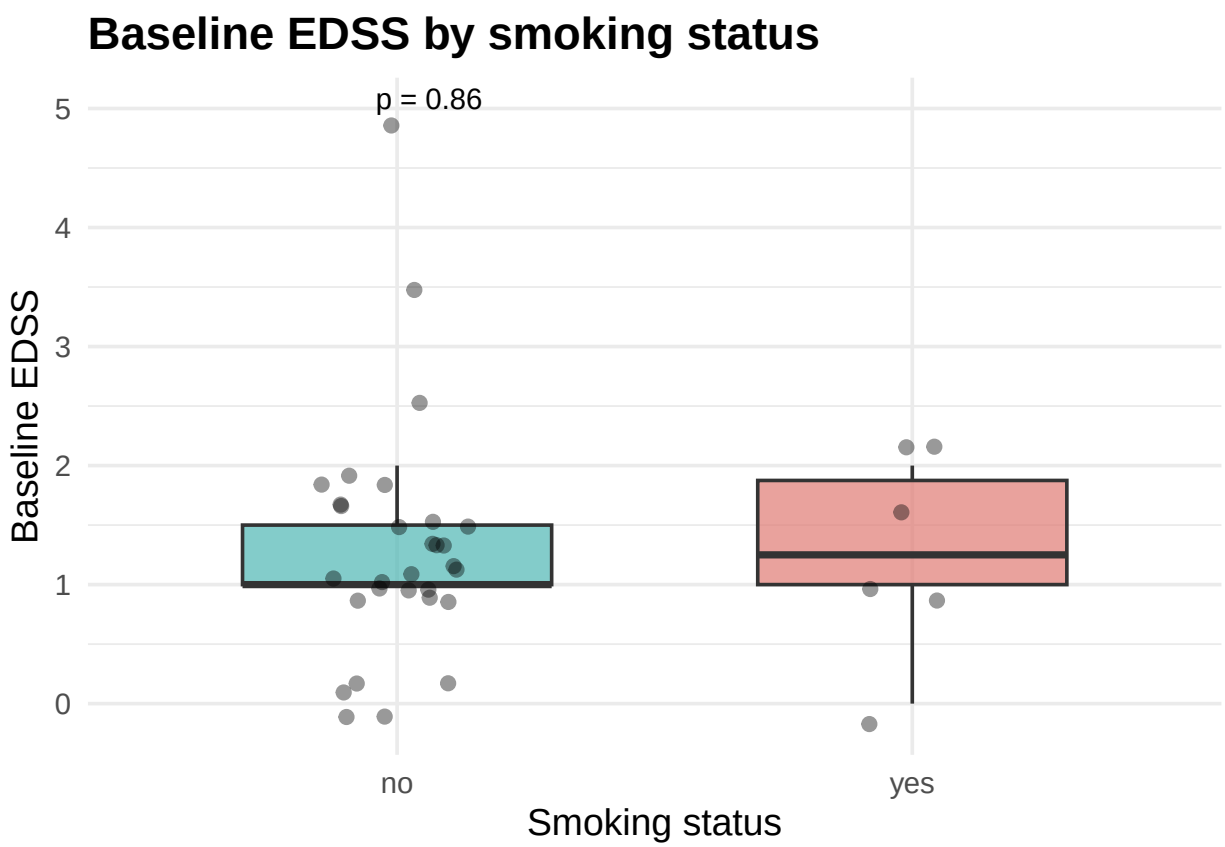
A tendency toward higher baseline EDSS values was observed in patients with a history of smoking compared to non-smokers. However, this difference did not reach statistical significance ($p = 0.06$).

```r
ggplot(merged %>% filter(!is.na(smoking_habit)),
  aes(x = smoking_habit, y = EDSS_T0, fill = smoking_habit)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, width = 0.6) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 2, color = "black") +
  stat_compare_means(method = "wilcox.test", label = "p.format", size = 4) +
  scale_fill_manual(values = c("yes" = "#E07A73", "no" = "#4DB6B3")) +
  labs(
    title = "Baseline EDSS by smoking status",
    x = "Smoking status",
    y = "Baseline EDSS"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold")
  )
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_compare_means()`).
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): non è
## possibile calcolare p-value esatto in presenza di ties
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  EDSS_T0 by smoking_habit
## W = 85.5, p-value = 0.8603
## alternative hypothesis: true location shift is not equal to 0
```

##Physiscal activity and baseline EDSS Baseline EDSS differed significantly across levels of physical activity ($p = 0.006$). Lower EDSS values were observed in more physically active patients, suggesting an inverse association between physical activity and disability, although causality cannot be inferred.

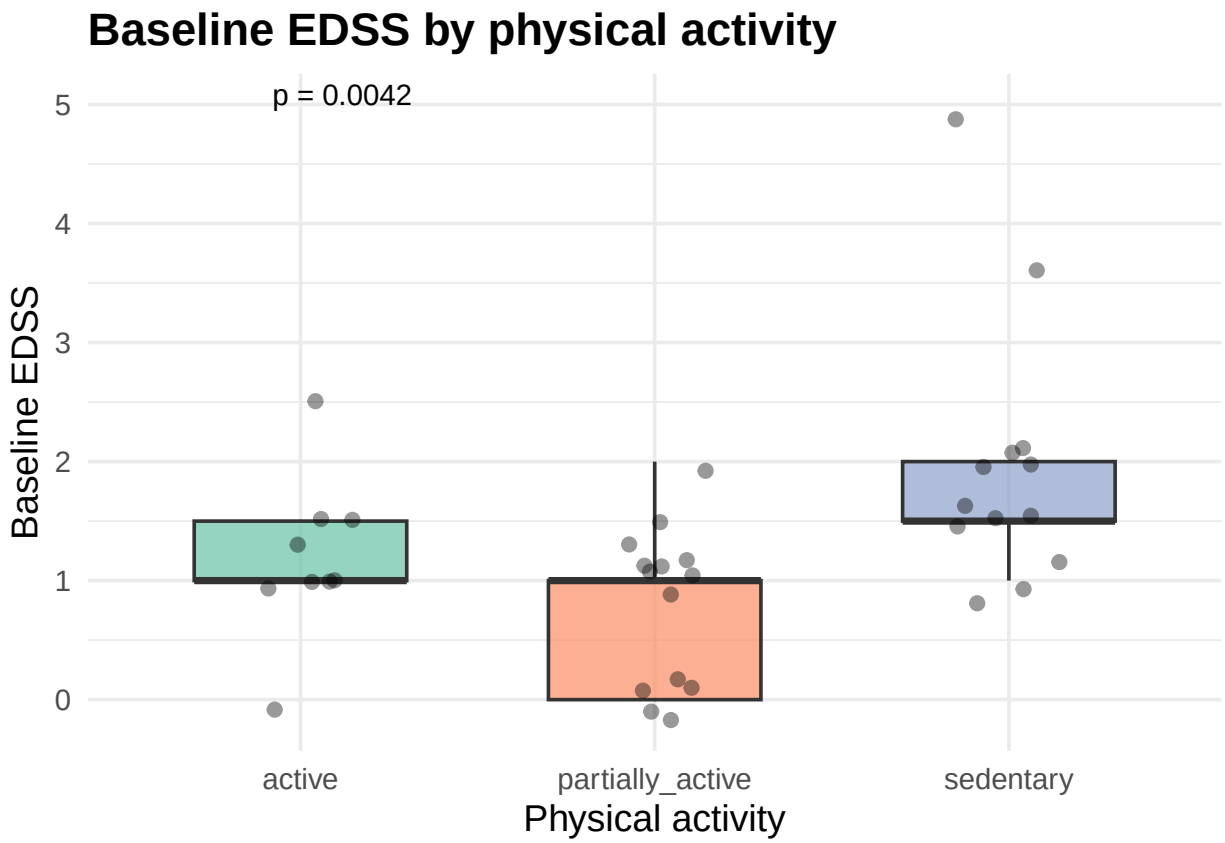
```
ggplot(merged %>% filter(!is.na(physical_activity)),
       aes(x = physical_activity, y = EDSS_T0, fill = physical_activity)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, width = 0.6) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 2, color = "black") +
  stat_compare_means(method = "kruskal.test", label = "p.format", size = 4) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Baseline EDSS by physical activity",
    x = "Physical activity",
    y = "Baseline EDSS"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold")

    )
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (‘stat_compare_means()‘).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_point()‘).
```




```
##
##  Kruskal-Wallis rank sum test
##
## data:  EDSS_T0 by physical_activity
## Kruskal-Wallis chi-squared = 10.929, df = 2, p-value = 0.004234
```

Conclusion

EDSS evolution over time showed variability between treatment groups without a clear overall progression. Baseline disability was associated with lesion burden and physical activity, while no clear associations were observed for other factors. These findings should be interpreted with caution due to the observational design.